ARYA DADHICH

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PROFESSIONAL SUMMARY

Data Engineer and AI/ML practitioner with hands-on experience designing end-to-end data pipelines, deploying computer vision solutions, and building analytical models that drive business decisions. Proficient in cloud-native data engineering (AWS), big data ecosystems (Spark, Hadoop), and machine learning frameworks (TensorFlow, PyTorch). Strong foundation in Business Analytics from an integrated MTech programme, complemented by leadership experience in competitive hackathons and large-scale student organisations.

EDUCATION

Integrated MTech Computer Science Engineering | **Vellore Institute of Technology** *2019 – 2024*

Specialisation: Business Analytics | **CGPA:** 8.12 / 10

Key Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Big Data Analytics, NoSQL Databases, Data Visualisation & Analysis, Statistical Analysis, DBMS, Information Retrieval, R-Studio

WORK EXPERIENCE

Data Engineer | **ZS Associates** *Pune, IN · Jan 2025 – Apr 2026*

- Designed and maintained scalable data pipelines spanning the full data lifecycle — Landing → Standardised → Normalised → MDM — handling source onboarding, schema evolution, transformations, and validation to ensure downstream readiness for MDM systems and analytics consumers.
- Performed impact analysis and data quality checks for new and existing data sources using SQL-based validation scripts and Python automation; identified and resolved missing, inconsistent, or incorrect attributes before they propagated downstream.
- Supported production deployments and environment migrations, including release validation, knowledge transfer documentation, and operational automation to reduce manual effort and improve pipeline reliability.
- Collaborated with cross-functional teams to translate business requirements into technical pipeline specifications, ensuring alignment between data engineering outputs and analytics/reporting needs.

Tech Stack: *Python, SQL (AWS Athena), AWS S3, AWS Glue, AWS EMR, Apache Airflow, Git, ETL Workflows*

AI/ML Intern | **Marico Limited** *Mumbai, Hybrid · Jan 2024 – Jul 2024*

Project 1: Share of Shelf — Modern Trade (Computer Vision)

Supervised Learning · Unsupervised Learning · Reinforcement Learning

- Engineered and deployed an advanced object detection model using Single Shot Detection (SSD) with ElasticNet and MobileNet backbones to accurately identify over 50 SKUs on modern trade retail shelves, enabling precise, automated product tracking at scale.
- Processed and annotated approximately 2,000 images per SKU category, building a robust end-to-end pipeline encompassing data ingestion, preprocessing, batch inference, and continuous model retraining cycles — significantly improving detection accuracy over successive iterations.
- Designed the full analytics pipeline and real-time dashboard that provided the Modern Trade (MT) team with live visibility into SKU placement and competitor shelf activity across retail locations, enabling strategic assortment and compliance decisions backed by data.
- Integrated large-scale distributed data processing using Spark and Hadoop (HDFS) to handle high-volume image metadata and inference results efficiently.

Tech Stack: *TensorFlow, Keras, PyTorch, Scikit-learn, Apache Spark, Hadoop, HDFS, Python, Visualisation Tools*

Project 2: Trade Offer Optimisation (Statistical Modelling)

- Engineered a linear regression model to quantify the correlation between trade promotional offers and sales volume, surfacing data-driven insights to optimise promotional strategy.
- Constructed and executed a multilevel regression model to evaluate the differential impact of offers across organisational hierarchies (channel, geography, SKU level), applying custom feature elimination techniques in Python to improve model parsimony and predictive accuracy.
- Partnered directly with the Trade Marketing team to translate model outputs into actionable spend allocation recommendations across channels, geographies, and SKUs — contributing to measurable improvements in promotional ROI.

Tech Stack: *Python, Scikit-learn, Linear Regression, Multilevel Regression, Feature Elimination, Data Analysis*

SELF-LEARNING PROJECTS

Detection of Renal Calculi (Kidney Stones)

Jul 2021 – Dec 2021

Medical Image Processing · Segmentation · Classification

- Led development of an advanced diagnostic tool for detecting renal calculi from medical images, incorporating pre-processing techniques including colour deconvolution to improve image clarity and downstream segmentation quality.
- Developed and fine-tuned segmentation algorithms through morphological reconstruction, achieving over 95% accuracy in feature extraction and classification of renal stones.
- **Outcome:** 20% reduction in diagnostic errors, significantly improving reliability for clinical application.

Tech Stack: *MATLAB, MATLAB Image Processing Toolbox*

Meme-Enhanced Hybrid Movie Recommender System with Cold Start Resolution

Dec 2022 – Mar 2023

Recommender Systems · Hybrid Filtering · Cold Start Problem

- Identified and addressed the cold start problem in collaborative filtering — where new users with no history receive poor recommendations — by innovatively incorporating profile-based memes and popular film dialogues as proxy preference signals to bootstrap personalisation from day one.
- Architected a hybrid recommendation engine combining content-based filtering (item features) and collaborative filtering (user-item interaction patterns), enabling nuanced and personalised recommendations across diverse user profiles.
- Built a robust, scalable dashboard using Python, Pandas, and Streamlit with an HTML/CSS front-end, integrating multiple visualisation libraries to deliver an intuitive user interface for exploring recommendations.

Tech Stack: *Python, Pandas, Streamlit, HTML/CSS, Tableau, Power BI, Plotly, ggplot2, Matplotlib, Seaborn*

Gesture-Based Upper-Body Recognition for Virtual Game Interfaces

Dec 2021 – Mar 2022

Computer Vision · Real-Time Recognition · Human-Computer Interaction

- Designed an intuitive gesture recognition system combining gesture spotting with real-time recognition using the MediaPipe pose estimation framework, enabling fluid, camera-based control of virtual game environments without physical input devices.
- Mapped upper-body skeletal landmarks to custom game control bindings using PyAutoGUI, creating a seamless and immersive user experience and demonstrating practical application of computer vision to human-computer interaction.
- Backed the solution with a multi-database architecture to store user gesture profiles and session data, supporting extensibility and future personalisation features.

Tech Stack: *Python, OpenCV, MediaPipe, PyAutoGUI, MySQL, MongoDB, Cassandra, Oracle GCP*

ACHIEVEMENTS & COMPETITIONS

Finalist — RAKATHON Hackathon | Rakuten

Chennai · Jul 2022

- Led a team of 4 members to the top 100 of the Rakuten RAKATHON, competing on the theme of Cloud technologies.

- Developed a dynamic video scaling solution powered by deep neural networks, optimising video quality adaptively based on network conditions and device constraints.

LEADERSHIP & EXTRACURRICULARS

General Secretary & Core Member | **Dramatics Club, VIT** *Chennai, IN · During B.Tech*

- Served as General Secretary of the Dramatics Club, leading a team of 20+ members across direction, screenwriting, and rehearsals for full-scale theatrical productions.
- Represented VIT at inter-collegiate theatre festivals at IITs and NITs across India, competing at a national level and building the club's external profile.
- Participated in every club event as a core creative member: contributing to script development, stage direction, and performance; while simultaneously managing logistics, scheduling, and team coordination.

High-Altitude Trekking *Multiple Expeditions*

- Completed multiple high-altitude treks reaching peaks till 15,000 ft, requiring sustained physical endurance, mental resilience, and disciplined preparation.

VOLUNTEERING & SOCIAL IMPACT

Volunteer | **Helphen India** *Tamil Nadu / Bengal · During COVID-19*

- Contributed to Helphen India's Period Sanity Drive, working to ensure menstrual hygiene access for underserved communities during COVID-19 lockdowns.
- When flood-affected communities in Bengal; cut off from aid due to the pandemic; reached out to Helphen India despite its operation only in Tamil Nadu, coordinated a cross-state relief effort: **collaborated with Tata Motors** to source and deliver first aid kits and food supplies to flood victims, funded entirely through a public donation drive.
- This initiative demonstrated resourcefulness, cross-organisational collaboration, and commitment to impact beyond organisational boundaries.

TECHNICAL SKILLS

Languages & Query: Python, SQL, R, MATLAB

Cloud & Data Engineering: AWS (S3, Glue, Athena, EMR), Apache Airflow, Apache Spark, Hadoop, HDFS, ETL Pipelines, Git

Machine Learning & AI: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, MediaPipe, Object Detection (SSD, MobileNet, ElasticNet)

Data & Visualisation: Pandas, Tableau, Power BI, Plotly, Matplotlib, Seaborn, ggplot2, Streamlit

Databases: MySQL, MongoDB, Cassandra, Oracle, NoSQL

Analytics Domains: Big Data Analytics, Predictive Analytics, Prescriptive Analytics, Statistical Modelling, Regression Analysis

Soft Skills: Leadership, Communication, Storytelling, Project Management, Stakeholder Management

CERTIFICATIONS

- Getting Started with Data Science Enterprise: IBM SkillsBuild
- Python for Machine Learning: Kaggle
- Pandas: Kaggle

PERSONAL INTERESTS

Theatre & Performing Arts · Reading · Yoga · High-Altitude Trekking